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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

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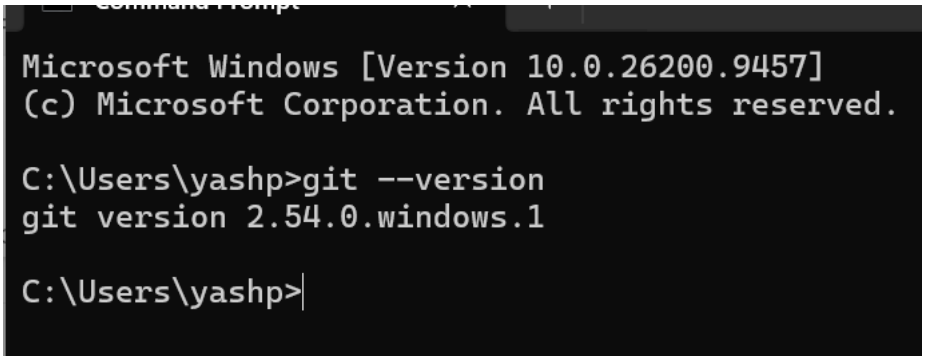
Re-accredited by NAAC with 'A' grade (3.58/4) | Awarded Category – I by UGC

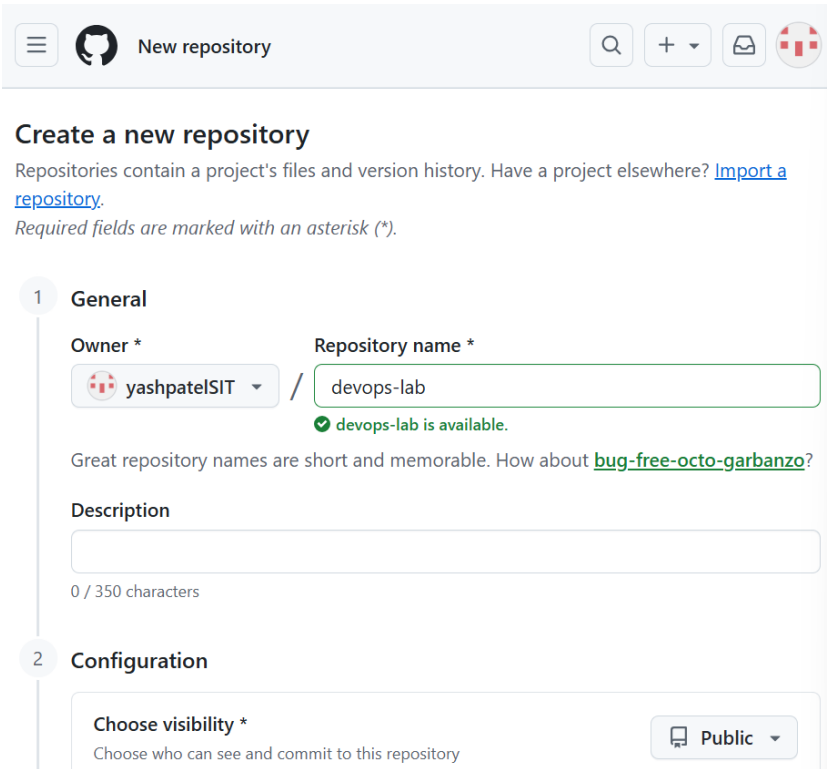
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

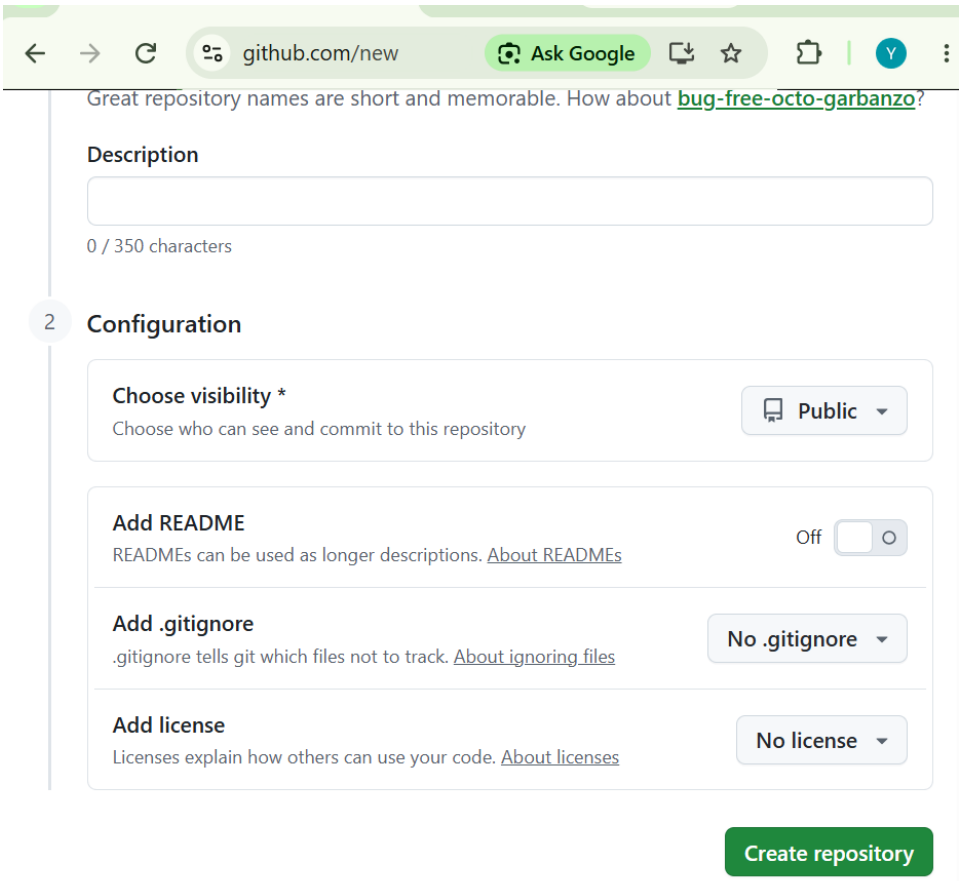
Assignment No. 01

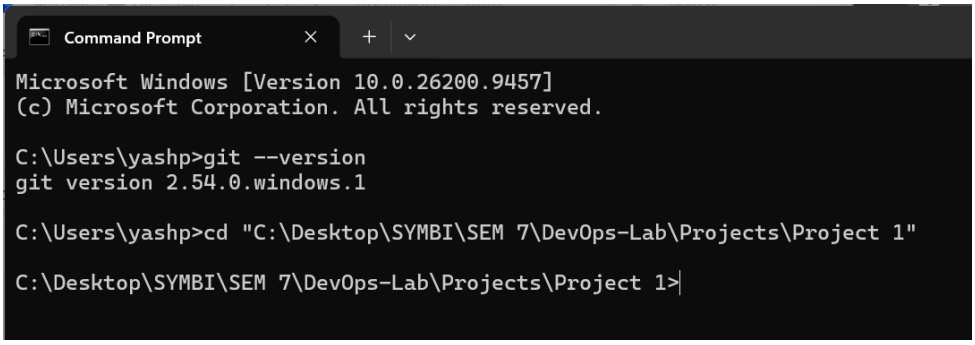
Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	**Yash Patel**
PRN No.	**23070122235**
Branch	CSE, Batch (2023-27), L4
Academic Year & Semester	23 – 27, Sem VII
Date of Performance	16th July, 2026
Title of Assignment:	Git Setup & Basic Workflow
Theory:	Git is a distributed version control system that helps developers track changes in source code, collaborate efficiently, and maintain different versions of a project. GitHub is a cloud-based platform that hosts Git repositories and enables collaboration. In this assignment, a GitHub repository was created and cloned to the local system. A README.md file was created, staged using git add, committed with git commit, and uploaded to GitHub using git push. This demonstrates the basic Git workflow for managing and synchronizing project files between a local machine and a remote repository. **Commands Used** 1. cd D:\CCodes – Navigated to the directory where the repository would be stored. 2. git clone https://github.com/riavinod10/devops-lab.git – Cloned the remote GitHub repository to the local machine. 3. cd devops-lab – Entered the cloned repository folder. 4. echo "# DevOps Lab" > README.md – Created the README.md file with initial content.

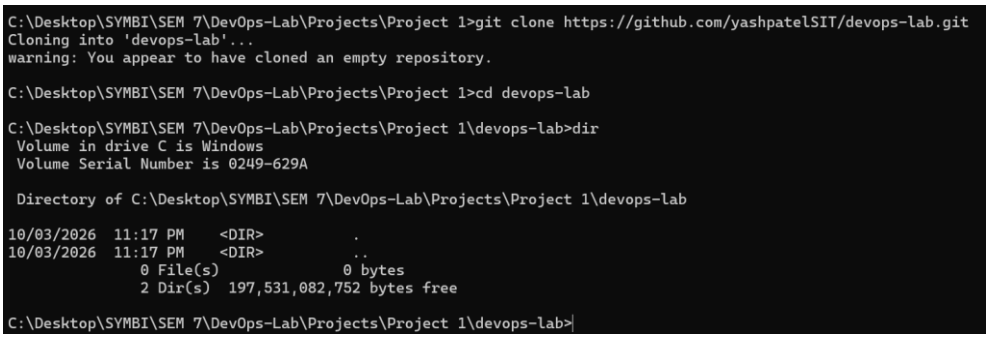
	5. git status – Checked the repository status to verify that README.md was listed as an untracked file. 6. git add README.md – Added the README.md file to the staging area. 7. git status – Verified that the file was staged and ready to be committed. 8. git commit -m "Add initial README.md file" – Created the first commit with a descriptive commit message. 9. git status – Confirmed that the working tree was clean after the commit. 10. git push origin main – Pushed the committed changes from the local repository to the remote GitHub repository.
Implementation:	**Step 1: Create a New Repository on GitHub** Empty Repository on GitHub — Created an empty repository on GitHub without initializing it with a README, .gitignore, or license file to practice the Git local-to-remote workflow. **Step 2: Clone the Repository** Repository Cloned- Cloned the remote GitHub repository to the local machine using git clone and navigated into the project directory. **Step 3: Create README.md** Created README.md- Created the README.md file with initial content and verified using git status that Git recognizes it as an untracked file. **Step 4: Stage the File** Staged README.md- Added the README.md file to the staging area using git add and confirmed it is ready to be committed. **Step 5: Commit the Changes** Committed Changes- Created the first commit with a descriptive commit message and verified that the working tree is clean. **Step 6: Push to GitHub** Pushed to Remote Repository: Uploaded the local commit to the remote GitHub repository using git push. **Step 7: Verify on GitHub** Repository Verified: Verified on GitHub that the README.md file and the latest commit are visible, confirming the push was successful. Assignment 1:


Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.
C:\Users\yashp>git --version
git version 2.54.0.windows.1
C:\Users\yashp>


New repository
Create a new repository
Repositories contain a project's files and version history. Have a project elsewhere? Import a repository.
Required fields are marked with an asterisk (*).
1
General
Owner *
Repository name *
yashpatelSIT
/
devops-lab
devops-lab is available.
Great repository names are short and memorable. How about bug-free-octo-garbanzo?
Description
0 / 350 characters
2
Configuration
Choose visibility *
Choose who can see and commit to this repository
Public


github.com/new
Ask Google
Great repository names are short and memorable. How about bug-free-octo-garbanzo?
Description
0 / 350 characters
2
Configuration
Choose visibility *
Choose who can see and commit to this repository
Public
Add README
READMEs can be used as longer descriptions. About READMEs
Off
Add .gitignore
.gitignore tells git which files not to track. About ignoring files
No .gitignore
Add license
Licenses explain how others can use your code. About licenses
No license
Create repository


Command Prompt
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.
C:\Users\yashp>git --version
git version 2.54.0.windows.1
C:\Users\yashp>cd "C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 1"
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 1>


C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 1>git clone https://github.com/yashpatelSIT/devops-lab.git
Cloning into 'devops-lab'...
warning: You appear to have cloned an empty repository.
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 1>cd devops-lab
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 1\devops-lab>dir
Volume in drive C is Windows
Volume Serial Number is 0249-629A
Directory of C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 1\devops-lab
10/03/2026 11:17 PM <DIR> .
10/03/2026 11:17 PM <DIR> ..
0 File(s) 0 bytes
2 Dir(s) 197,531,082,752 bytes free
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 1\devops-lab>

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 1\devops-lab>git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.md


C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 1\devops-lab>git commit -m "Add initial README.md fi
le"
[main (root-commit) 88b5cbe] Add initial README.md file
 1 file changed, 1 insertion(+)
 create mode 100644 README.md

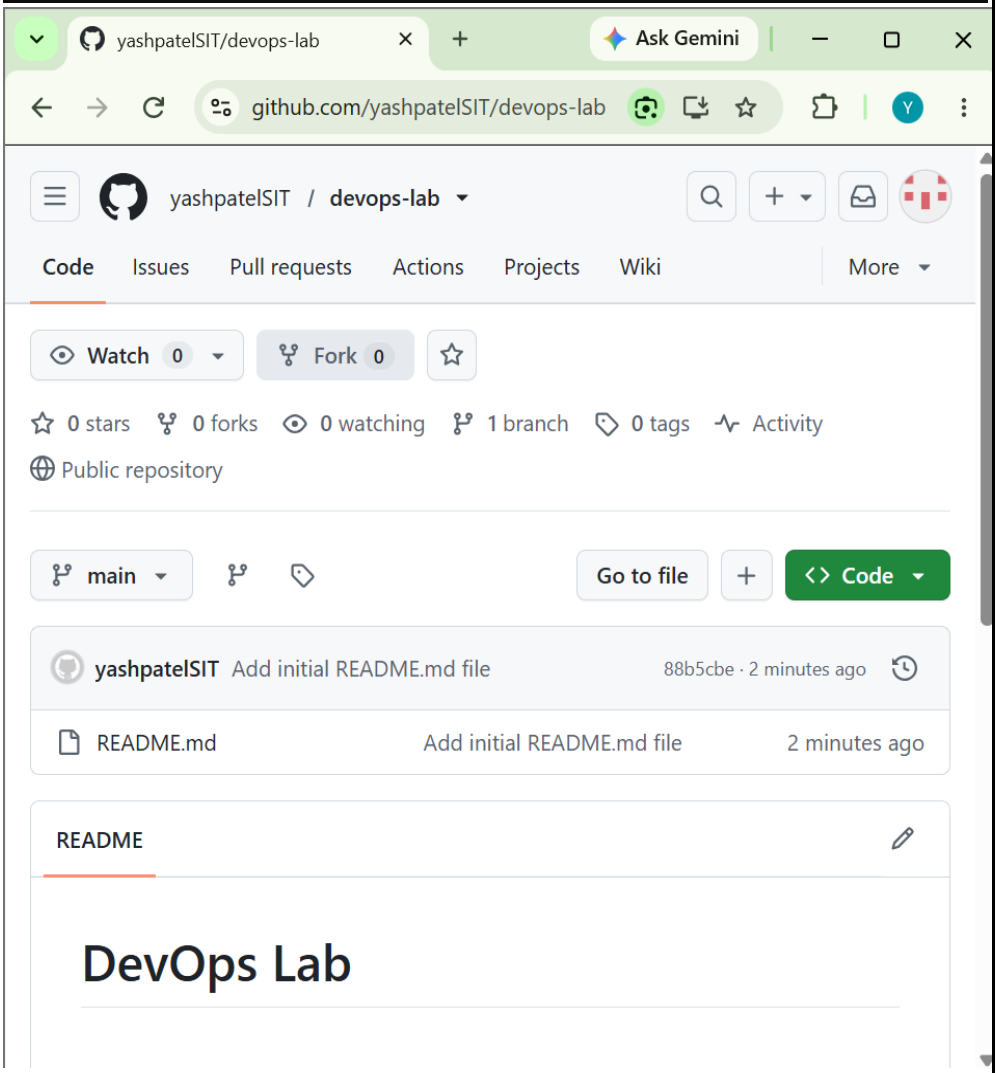
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 1\devops-lab>git status
On branch main
Your branch is based on 'origin/main', but the upstream is gone.
  (use "git branch --unset-upstream" to fixup)

nothing to commit, working tree clean

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 1\devops-lab>
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 1\devops-lab>git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 245 bytes | 61.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/yashpatelSIT/devops-lab.git
 * [new branch]      main -> main

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 1\devops-lab>
```



Subject Faculty Name: Prof. Arijit Dutta